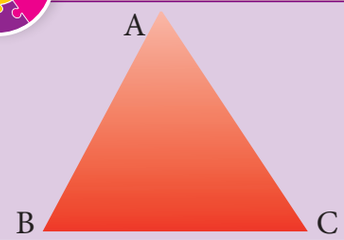
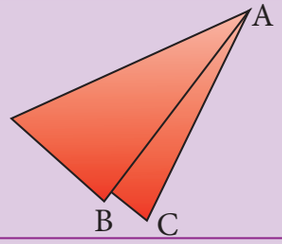
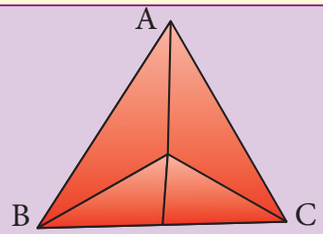
Activity

		
1. Consider a paper cut-out of a triangle. Name it, say ABC.	2. Fold the triangle so that the opposite sides meet and contain the vertex. Repeat the same to find angle bisectors of other two angles also.	3. Trace all of the folds. Do the angle bisectors pass through the same point?

Now you can repeat this activity for an obtuse-angled triangle and a right angled triangle. What is the conclusion? Do the angle bisectors pass through the same point in all the cases? Yes, we see that,

The three angle bisectors of any triangle are concurrent.

5.9.1 Incentre

The point of concurrence of the three angle bisectors of a triangle is called as its **incentre,** denoted by the letter **I**.

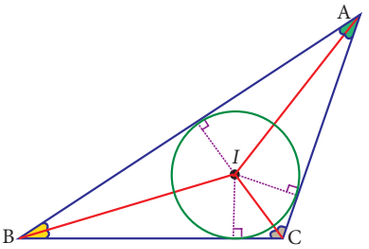


Fig. 5.38

Why should it be called so? Because one can draw a circle inside of the triangle so that it touches all three sides internally, with centre at the point of concurrence of the angle bisectors. The lengths of a perpendicular line drawn from incentre to each side is found to be same. Thus, the incentre is equidistant from the sides of the triangle.

Example 5.20

Identify the incentre of the triangle PQR.

Solution:

Incentre is the point of intersection of angle bisectors.

Here, PM and QN are angle bisectors of ∠P and ∠Q respectively, intersecting at B.

So, the incentre of the triangle PQR is B.

(Can A and C be the incentre of ΔPQR? Why?)

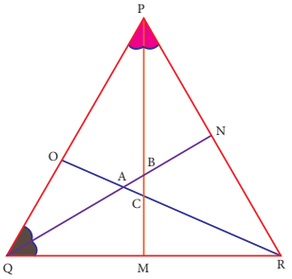


Fig. 5.39

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